VeriSelf: Neutralizing AI Threats to Personal Privacy

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Abstract:
Personal photographs shared online can be copied, misused for impersonation, or altered without the owner permission, leaving individuals with limited ways to investigate suspicious content and organize supporting evidence. VeriSelf is a full-stack web application developed to support personal digital identity protection through image analysis, identity verification, public-web monitoring, and evidence management. The system uses React and Vite to provide the user interface, FastAPI to handle backend operations, and SQLAlchemy with SQLite to manage application records. For identity verification, it employs MTCNN for face detection and an InceptionResnetV1 model pretrained on VGGFace2 to generate facial embeddings for comparison. Perceptual hashing helps identify visually similar images, while image and video analysis modules examine characteristics such as noise inconsistency, edge density, texture variation, and colour-channel differences to identify potential manipulation indicators. The monitoring module scans a user-specified public webpage for candidate images, and the evidence module organizes investigation findings for further review. Additional functions support photograph protection, risk assessment, alerts, and the preparation of draft legal-response notices. By bringing these capabilities together, VeriSelf provides a structured workflow for examining suspected digital identity misuse and documenting relevant findings. Its results require human review because facial similarity does not guarantee identity, perceptual matching cannot establish ownership, and heuristic forensic indicators alone cannot confirm image or video manipulation.

Key words: Digital Identity Protection, Face Detection, Facial Embeddings, Perceptual Hashing, Public-Web Monitoring, Image Forensics, Video Analysis, Risk Assessment, Digital Evidence Management

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